



SCHOOL OF COMPUTING AND ENGINEERING SCIENCES (SCES)
Bachelor Of Science In Informatics And Computer Science

ICS 1103 - 3D Operations Research
Class work June 4, 2026

Instructions

1. Read the course notes on **Duality Theory and Sensitivity Analysis** carefully before attempting this assignment.
 2. Answer **BOTH** questions.
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QUESTION ONE

[15 Marks]

A transport company operates a fleet of trucks and vans to fulfil daily cargo deliveries. Two vehicle types are available, **Trucks** and **Vans**.

- Each **Truck** trip costs **KES 5** and carries **2 tonnes** of northbound cargo and **3 tonnes** of southbound cargo.
- Each **Van** trip costs **KES 4** and carries **3 tonnes** of northbound cargo and **1 tonne** of southbound cargo.
- The daily schedule must move **at least 17 tonnes** northbound and **at least 15 tonnes** southbound.

Let y_1 be the number of Truck trips and y_2 the number of Van trips per day. The linear programme is:

$$\text{Minimise } W = 5y_1 + 4y_2$$

$$2y_1 + 3y_2 \geq 17, \quad 3y_1 + y_2 \geq 15, \quad y_1, y_2 \geq 0$$

- (a) Write the coefficient matrix of the primal problem and transpose it to obtain the **dual maximisation problem**. State the dual objective function and all dual constraints clearly. [4]
- (b) Introduce slack variables and write the **initial simplex tableau** for the dual problem. Label all columns and rows. [3]

- (c) Apply the simplex method to solve the dual problem. Show every tableau and indicate the pivot element at each stage. State the maximum value of Z and the values of x_1 and x_2 . [5]
- (d) Using the final tableau from (c), state:
- (i) The optimum number of Truck trips y_1 and Van trips y_2 .
 - (ii) The minimum daily transport cost W .
 - (iii) The shadow values for northbound and southbound cargo, and explain what each shadow value means for the transport manager.

[3]

QUESTION TWO**[15 Marks]**

A cleaning company must prepare a disinfectant solution each day using two chemical products, **Detergent I** and **Detergent II**.

- Each litre of **Detergent I** costs **KES 2** and provides **2 units** of disinfectant agent and **1 unit** of degreaser.
- Each litre of **Detergent II** costs **KES 3** and provides **1 unit** of disinfectant agent and **3 units** of degreaser.
- The daily solution must contain **at least 11 units** of disinfectant agent and **at least 8 units** of degreaser.

Let y_1 be the litres of Detergent I and y_2 the litres of Detergent II used per day. The linear programme is:

$$\text{Minimise } W = 2y_1 + 3y_2$$

$$2y_1 + y_2 \geq 11, \quad y_1 + 3y_2 \geq 8, \quad y_1, y_2 \geq 0$$

- (a) Using the four-step procedure from your notes, write the primal coefficient matrix, transpose it, and state the **dual maximisation problem** fully. [3]
- (b) Set up the **initial simplex tableau** for the dual problem. [2]
- (c) Apply the simplex method to solve the dual. Show every tableau and indicate all pivot elements. State the maximum value of Z and the optimum values of x_1 and x_2 . [5]
- (d) From the final tableau, read off the solution to the **original minimisation problem**. State the optimum values of y_1 and y_2 and the minimum daily chemical cost W . Confirm that the minimum value of W equals the maximum value of Z . [3]
- (e) A health inspector informs the company that the minimum requirement for disinfectant agent will increase by **5 units** next month. Using the shadow value for the disinfectant constraint from your final tableau, estimate the **increase in the minimum daily cost** and interpret this for the company manager. [2]